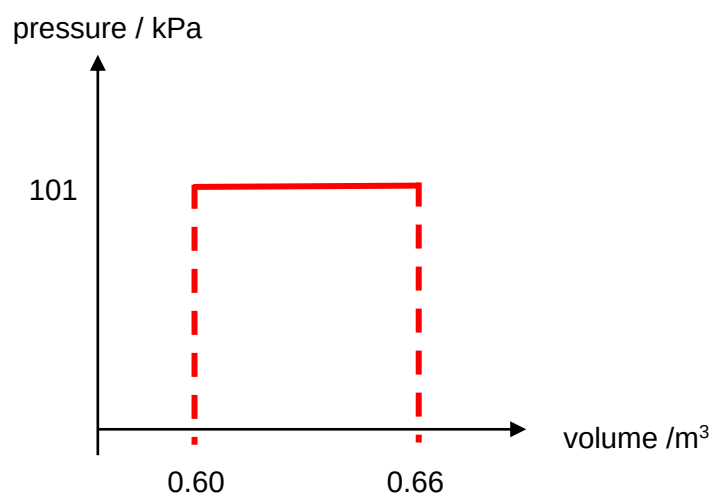


- 3 (a) (i) When the gas is heated, the molecules gains kinetic energy (or speed), leading to a larger change in momentum per collision with the container per unit time. This leads to a larger force on the walls of container and piston. B1

This leads to an increase in pressure in the container. To keep the pressure constant, the piston moves up that leads to an increase in volume of gas until the rate of change of momentum of gas molecules with the walls of the container decreases to the original level. B1

(ii)

B2



$$\frac{0.60}{28 + 273.15} = \frac{V_2}{57 + 273.15}$$
$$V_2 = 0.66 \text{ m}^3$$

1 mark for correct graph

1 mark for correct showing correct values on both axes

(iii) Energy from heater = (24)(15)(60) = 21600 J

Using  $\Delta U = Q + W$ ,

$$Q = -7000 + 21600 = 14600 \text{ J}$$

B1

$$W = 101000(0.66 - 0.60) = 6060 \text{ J}$$

C1

$$\Delta U = 14600 - 6060 = 8540 \text{ J}$$

A1

Accept slight variation in values if using raw values from (ii)

Max of 1 mark if students using alternative methods

(b) (i) Using  $pV = \frac{1}{3}Nm\langle c^2 \rangle$  and  $pV = NkT$ ,

$$\frac{2}{3}N\left(\frac{1}{2}m\langle c^2 \rangle\right) = NkT$$

B1

$$E = \frac{1}{2}m\langle c^2 \rangle = \frac{3}{2}kT$$

(ii) Using  $\frac{1}{2}m\langle c^2 \rangle = \frac{3}{2}kT$ ,  $k = \frac{R}{N_A}$  and  $M = N_A m$

$$c_{rms} = \sqrt{\langle c^2 \rangle} = \sqrt{\frac{3kT}{m}} = \sqrt{\frac{3RT}{M}} = \sqrt{\frac{3(8.31)(57 + 273.15)}{4.0 \times 10^{-3}}} = 1.43 \times 10^3 \text{ m s}^{-1}$$

C1

A1

4 (a) A progressive wave is a wave in which energy is carried from B1